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# SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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## ABSTRACT

Mixture of Experts (MoE) models are central to scaling large language models, but their empirical representational differences from dense networks remain under-specified. Dense toy models are known to use *superposition* to represent more features than dimensions, with phase behavior governed by feature sparsity and feature importance. We show that the same quantitative framing does not explain MoEs: across our controlled comparisons, neither feature sparsity nor feature importance produces the discontinuous phase changes seen in dense models, while network sparsity (the ratio of active to total experts) more directly tracks representational behavior. We introduce metrics for comparing superposition across experts and find a consistent pattern: as network sparsity increases, MoE representations become more *monosemantic*. Under equal-parameter comparisons, MoEs reduce feature interference relative to dense baselines while maintaining comparable reconstruction loss, with observed loss gaps of roughly 0.03–0.08 at fixed sparsity. We further define expert specialization by monosemantic feature representation rather than load balancing alone, and show that appropriately initialized experts align with coherent feature combinations. Together, these results frame network sparsity as an empirically measurable route toward more interpretable MoE representations without a large practical loss penalty.

## 1 INTRODUCTION

Mixture of Experts (MoEs) are now common in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), because they deliver favorable compute–performance trade-offs (Shazeer et al., 2017; Fedus et al., 2022). Follow-up work improves routing and optimization through mechanisms such as Expert-Choice routing and ST-MoE (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical studies also show that learnable routers can recover latent cluster structure in data, giving a quantitative basis for expert specialization (Dikkala et al., 2023). Yet the same empirical clarity is missing at the mechanistic level: we still do not know which representational quantities distinguish MoEs from dense models.

Interpretability-oriented MoE approaches have already produced useful architectural baselines. Yang et al. (2025b) proposes MoE-X, which combines sparsity-aware routing with wide, ReLU-based experts to reduce polysemanticity, while Park et al. (2025) introduce MONET, where expert count is scaled to support capability editing through expert activation. These systems make expert behavior more controllable, but their evidence primarily concerns architectural interventions rather than direct comparisons of feature geometry. In particular, the field lacks measurements of how MoE experts change superposition, whether those changes are consistent across sparsity and importance conditions, and whether specialization emerges without extra regularization. Mu & Lin (2025) therefore identify mechanistic interpretability as a key open challenge for MoEs.

A central quantitative obstacle in interpreting neural networks is superposition: models can represent more features than available dimensions. This packing increases representational

capacity, especially for sparse features, but it also raises feature interference and makes individual neurons polysemantic.

MoE architectures add a second sparsity axis to this comparison: network sparsity. Dense models activate all neurons for every input, whereas MoEs activate only a fraction of their total parameters (Shazeer et al., 2017). This creates a measurable contrast in representational strategy: dense models exploit feature sparsity by packing many features into shared dimensions, while MoEs can use conditional computation to allocate expert capacity to narrower feature combinations.

We ask three comparison-driven questions. First, under matched parameter budgets, do MoEs exhibit lower superposition than dense baselines? Second, when feature sparsity and relative feature importance are varied, do MoE experts show the same discrete phase changes reported for dense toy models? Third, can expert specialization be measured through feature representation and routing alignment rather than through load-balancing statistics alone?

We study these questions by extending the toy-model framework of Elhage et al. (2022) to MoEs and by keeping each comparison explicit in terms of architecture, metric, sparsity condition, and baseline. Our results show that: (1) unlike dense models, MoEs do not exhibit sharp phase changes and instead show more continuous transitions as network sparsity increases; (2) MoEs consistently exhibit greater monosemanticity (less superposition) than dense models with equivalent active or total parameter comparisons, with individual experts representing features more cleanly; (3) expert specialization is better captured by monosemantic feature representation and conditional expert usage than by aggregate load balancing alone, with experts organizing around coherent feature combinations when initialization supports that alignment.

## 2 RELATED WORK

**Superposition and Feature Representations.** The Linear Representation Hypothesis treats concepts as directions in activation space (Park et al., 2024), but empirical feature counts can exceed the number of available dimensions. Elhage et al. (2022) formalized this mismatch as superposition, showing in dense models that non-orthogonal feature packing increases capacity while also increasing polysemanticity. Although superposition is theoretically efficient (Scherlis et al., 2025), it shifts interpretability burden to post-hoc disentanglement methods such as Sparse Autoencoders (Bricken et al., 2023). Recent work quantifies how correlations in data affect superposition (Prieto et al., 2025) and how interference patterns arise and can be mitigated (Gurnee et al., 2023). We extend this quantitative program to MoEs and show that network sparsity, not feature sparsity alone, is the comparison variable that best explains the observed representational patterns.

**Interpretability of MoEs.** MoEs are a leading scaling strategy for large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), yet their internal representations have been measured less directly than dense representations. Existing analyses emphasize aggregate phenomena such as routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). Recent interpretability-focused systems introduce stronger architectural constraints, including pre-defined concept routing (Yang et al., 2025b) and vocabulary-scale expert counts (Park et al., 2025). These baselines clarify how expert behavior can be shaped, but they do not isolate the intrinsic feature-geometry effect of experts. We therefore compare dense and MoE representations using superposition metrics, report the consistent monosemanticity advantage of MoEs, and define expert specialization through feature-level representation and usage patterns.

## 3 BACKGROUND

A primary goal of Mechanistic Interpretability is to reverse engineer neural networks by decomposing model representations into human-interpretable concepts, or ‘features’. In the

linear-features framing used for quantitative comparison, a hidden state  $h$  is written as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where  $\vec{f}_i$  denotes the direction for feature  $i$ ,  $\alpha_i$  is its activation strength (roughly, the degree to which feature  $i$  is present in the input), and  $F$  is the full set of represented features.

The *superposition hypothesis* states that models can represent more features than dimensions, i.e.  $|F| > m$  for  $h \in \mathbb{R}^m$  (Elhage et al., 2022). Achieving this inequality requires feature vectors in  $F$  to be packed non-orthogonally. When two features have nonzero interference,  $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$ , we say they are in *superposition*. This trade-off is empirically useful when features are sparsely active, but it reduces interpretability because  $\alpha_i$  can include activation caused by other features rather than by feature  $i$  itself.

Monosemanticity is usually defined at the neuron level: a neuron is monosemantic when its activation cleanly tracks a single  $\alpha_i$  and the feature is basis-aligned. Orthogonal features can still be distributed across neurons, so low interference and strict basis alignment are not identical. Our measurements focus on the former quantity: reducing superposition. Thus, when we say that models, experts, or features are “more monosemantic,” we mean that their measured feature interference is lower.

## 4 DEMONSTRATING SUPERPOSITION

MoEs are often described as compositions of dense subnetworks, but that description leaves open whether each expert uses the same feature-packing strategy as a dense model. We begin with a direct empirical comparison of superposition: for matched toy settings, we measure which features are represented, how strongly they are represented, and how much they interfere with other features.

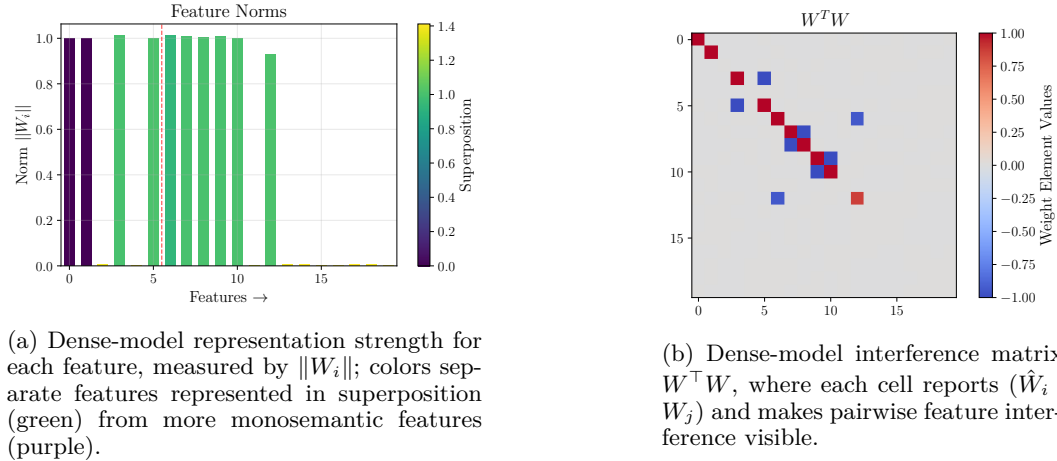


Figure 1: Dense baseline for feature representation and superposition with  $n = 20$  features and  $m = 6$  hidden dimensions, importance  $I = 0.7^i$ , and uniform feature density  $(1 - S) = 0.1$ . Bar heights show representation strength, while color encodes the superposition score  $\sum_j (\hat{W}_i \cdot W_j)^2$ .

### 4.1 EXPERIMENTAL SETUP

Our experimental goal is to compare how dense and MoE models compress and reconstruct the same high-dimensional input vector,  $x \in \mathbb{R}^n$ , through lower-dimensional representations  $h \in \mathbb{R}^m$ . This setup extends the framework of Elhage et al. (2022) to MoEs while preserving the quantities needed for side-by-side measurement of reconstruction loss, representation strength, and interference.

**The input distribution.** The input vector  $x$  represents activations in an idealized disentangled model, where each dimension  $x_i$  is a distinct independent feature and therefore begins as a perfectly neuron-aligned, monosemantic representation. We sample  $x_i \sim U(0, 1)$ , with feature sparsity controlled by  $S \in [0, 1)$  such that  $P(x_i = 0) = S$ . Because features need not contribute equally to reconstruction quality, each feature  $x_i$  receives a scalar importance weight  $I_i$  in the loss. To isolate phase behavior for one comparison feature, we scale the last feature by  $r \in \mathbb{R}^+$ , giving  $I = (1, 1, \dots, r)$ . Together,  $S$ ,  $I_i$ , and  $r$  define the evaluation condition: how often features are active and how costly each feature is to reconstruct.

**Model Architecture.** The MoE has  $E$  experts, and expert  $e$  is parameterized by a weight matrix  $W^e \in \mathbb{R}^{m \times n}$  and bias  $b^e \in \mathbb{R}^n$ . A learned router,  $g(x) = \text{softmax}(W^r x)$  with  $W^r \in \mathbb{R}^{E \times n}$ , assigns each input to the top- $k$  experts. Each active expert projects the input into a lower-dimensional hidden state  $h^e = W^e x$  and produces an expert-level reconstruction  $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ <sup>1</sup>. The final model output is the weighted sum over the active experts,  $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$ , where  $w_e$  are the renormalized gating weights. This notation makes the comparison condition explicit: dense models use one global representation, while MoEs distribute the same reconstruction objective across routed expert matrices.

All models are trained with an importance-weighted L2 reconstruction loss,  $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$ . In Section 4, where the comparison varies feature sparsity and last-feature importance over many cells, we add the standard auxiliary load-balancing loss (Fedus et al., 2022),  $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$ . Here,  $N$  is the total number of experts,  $f_e$  is the fraction of batch samples routed to expert  $e$ ,  $P_e$  is the average gating probability assigned to expert  $e$ , and  $\alpha = 0.01$  sets the penalty strength. We omit auxiliary load balancing in Section 4 so that the dense-MoE superposition comparison reflects the intrinsic architectural bias of routed experts rather than an explicit usage penalty.

## 4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we measure two quantities for each feature: *representation strength* and *interference*. Representation strength is the norm of the feature weight vector in expert  $e$ ,  $\|W_i^e\|$ ; values near one indicate that feature  $i$  is fully represented in expert  $e$ , while values near zero indicate that it is not learned there. Interference is visualized through the Gram matrix  $W^\top W$ , whose off-diagonal entries give pairwise feature interference within the relevant dense model or expert.

Figures 1a and 2a show that, under equal total hidden dimensions ( $m_{\text{dense}} = \sum m_{\text{experts}} = 6$ ), the dense model and MoE represent a comparable number of features: 10 in the dense model versus 8 in the MoE. The key difference is not feature count alone but interference structure. The MoE still shows local superposition inside individual experts, for example in Expert 0 of Figure 2b, but its global interference is partitioned by routing. In the dense model (Figure 1b), any represented feature can interfere with any other represented feature; in the MoE, features routed to different experts have zero cross-expert interference. Thus, at nearly comparable feature counts, the MoE changes the practical significance of superposition by localizing rather than globalizing interference.

**Expert Feature Dimensionality.** To quantify how MoEs allocate limited representational capacity relative to dense models, we use feature dimensionality, the “fraction of a dimension” assigned to a specific feature (Elhage et al., 2022). For feature  $i$  in expert  $e$ , dimensionality is

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

<sup>1</sup>The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where  $x_i \in [0, 1]$  when nonzero. Furthermore, the off-diagonal terms in  $(W^e)^\top W^e$  create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

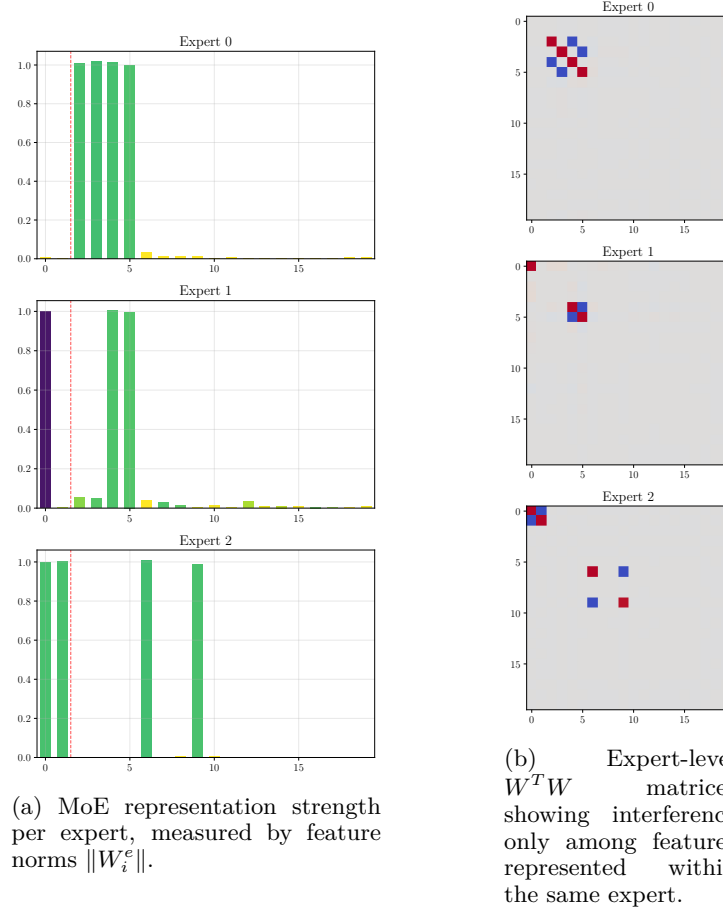


Figure 2: MoE feature representation and superposition under the same input condition as the dense baseline:  $n = 20$  features, 3 total experts,  $m = 2$  hidden dimensions per expert, top- $k = 1$  routing, importance  $I = 0.7^i$ , and feature density  $1 - S = 0.1$ . Compared with Figure 1, interference is localized within experts rather than shared globally.

$D_i^e$  ranges from zero, indicating that the feature is not learned in that expert, to one, indicating a monosemantic representation. Aggregating this quantity over features and experts gives the MoE capacity measure  $D = \sum_e \sum_{i=1}^n D_i^e$ .

**Efficient Packing.** A representation is “efficiently packed” when the summed feature dimensionality approximately equals the number of embedding dimensions,  $\sum_{i=1}^n D_i^e \approx m$  (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For an MoE, the analogous total-parameter comparison is  $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$ . Empirically, both dense and MoE models satisfy this constraint, so the central quantitative difference is not packing efficiency per parameter. Instead, the difference is how that packed capacity is distributed across experts and how much feature interference remains within each local representation.

**Features per Dimension & Network Sparsity.** Because dense and MoE models both efficiently pack features, we compare them using features per dimension, a direct measure of how much superposition remains after accounting for capacity. Values greater than one mean that the model represents more features than dimensions and therefore uses superposition. This metric also lets us isolate the effect of expert count under different feature-sparsity conditions. For an MoE, we define features per dimension as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

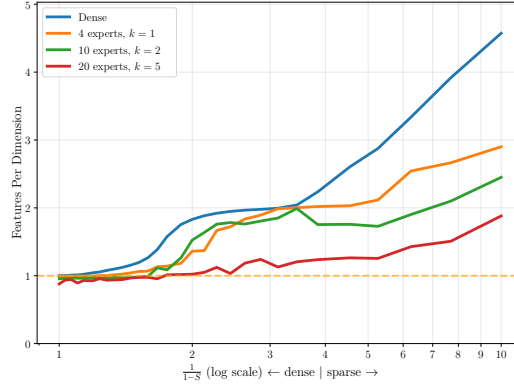


Figure 3: Features per dimension versus inverse feature density ( $\frac{1}{1-S}$ ) for equal-total-parameter **dense** and **MoE** architectures with uniform feature importance ( $I_i = 1.0$ ). The dense baseline ( $n = 100$ ,  $m = 20$ ) shows the highest features-per-dimension values, i.e. the most superposition. Increasing expert count monotonically reduces this metric across the MoE comparisons: 4 experts with  $m = 5$ ,  $k = 1$  (orange); 10 experts with  $m = 2$ ,  $k = 2$  (green); and 20 experts with  $m = 1$ ,  $k = 5$  (red). All models have equal total parameters and similar  $k/E$ ; the dashed line at 1.0 marks the monosemantic limit.

where  $\|W\|_F^2$  is the Frobenius norm and  $p_e$  is the expected usage probability of expert  $e$  over a batch, computed as the average renormalized gating weight after top- $k$  routing.

Under *equal total parameter* comparisons, the dense baseline has higher features per dimension than every MoE in Figure 3, meaning it uses more superposition. This indicates that the dense model represents more learned features than total hidden dimensions ( $n_{\text{learned}} > m_{\text{total}}$ ), while the MoE curves stay closer to the monosemantic limit ( $n_{\text{learned}} \approx m_{\text{total}}$ ). Holding total parameters and the ratio  $k/E$  roughly fixed, increasing the number of experts reduces features per dimension, so the empirical margin moves consistently in the same direction: more experts correspond to less superposition and more monosemantic feature representations. The dense model’s additional superposition yields lower reconstruction loss in Figure 6 of Appendix A.1, with fixed-sparsity loss differences of approximately 0.03–0.08. However, the practical gap narrows as expert count increases, and the higher-expert MoEs remain consistently comparable while providing substantially cleaner feature interference patterns. Appendix A.2 gives the corresponding intuition.

## 5 PHASE CHANGE

The preceding results show that MoEs and dense models can learn similar numbers of features, but MoEs distribute those features across experts with lower measured interference. This suggests that network sparsity changes the allocation pattern rather than simply changing the feature count. We therefore test whether input-level quantities that explain dense phase diagrams—feature sparsity and relative feature importance—produce the same phase-change structure in MoE experts.

Each model has a finite set of qualitative representational outcomes for a feature: it can ignore the feature, represent it in superposition, or represent it monosemantically. A phase change occurs when the boundary between these outcomes is sharp as a function of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between these internal feature representations (Elhage et al., 2022). By varying feature sparsity and relative importance, dense models move between ignorance, superposition, and monosemanticity; for example, higher feature sparsity can make superposition more attractive. We use the same axes for MoE experts so that any differences in the phase diagrams can be attributed to architecture and network sparsity rather than to a changed evaluation condition.

**Setup.** We use the setup in Section 4.1, adding load-balancing loss only to avoid specialization collapse in settings where routers and experts otherwise fall into local minima that ignore some experts. We evaluate three top- $k = 1$  architectures: (A)  $n=2, m=1$ ; (B)  $n=3, m=1$ ; and (C)  $n=3, m=2$ . For each architecture, we report expert-specific phase diagrams over feature sparsity and last-feature relative importance while increasing the total number of experts ( $E$ ) up to the number of input feature dimensions ( $n$ ).

For these phase-change comparisons, we fix active parameters rather than total parameters, so  $m_{dense} = km_{moe}$  when router parameters are ignored. This choice isolates differences within an architecture family; otherwise, shifts in the diagrams could reflect changed active capacity rather than changed network sparsity. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2, but the B models have only one hidden dimension ( $m = 1$ ) to encode three input features, whereas C.1/1 has two hidden dimensions ( $m = 2$ ). We therefore interpret each panel relative to its own capacity condition.

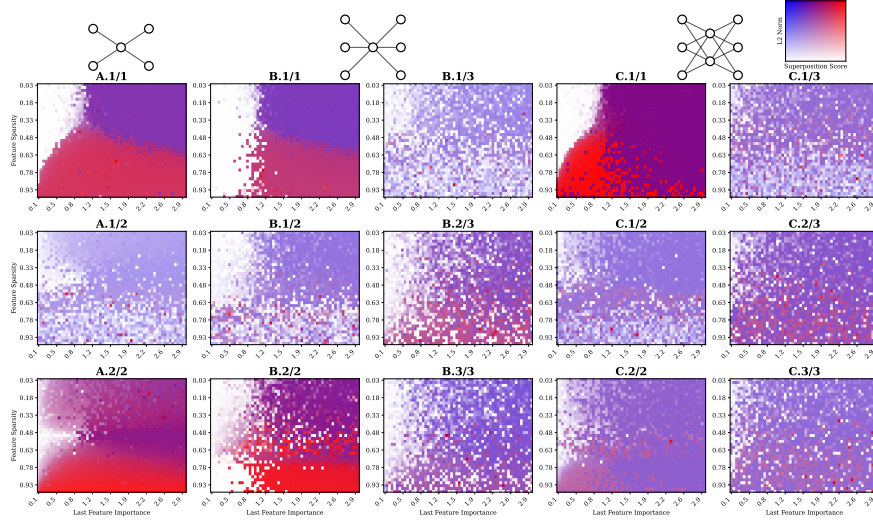


Figure 4: Phase diagrams for the last input feature under matched sparsity-importance sweeps. For each expert and feature, color summarizes whether the feature is ignored (white), represented monosemantically (blue-purple), or represented in superposition (red), using joint feature norm ( $\|W_n\|^2$ ) and superposition score ( $\sum_{j < n} (\hat{W}_n \cdot W_j)^2$ ). Columns vary feature sparsity  $S \in [0.1, 1]$  and rows vary relative last-feature importance  $I_n \in [0.1, 3]$ , where  $n$  denotes the last feature among  $n$  total features. For each cell, we train ten models with load-balancing loss and plot the lowest-loss run. A low L2 norm ( $\|W_n\|$ ) is white; otherwise, low superposition is blue-purple and high superposition is red. Cell  $(i, j)$  in subfigure X.e/E denotes expert  $e$  of  $E$  total experts for architecture X, last-feature importance  $I_n = i$ , and sparsity  $S = j$ ; X.1/1 is the dense baseline.

**Results and Takeaways.** The dense single-expert baselines show clear phase changes in every architecture (Figure 4.X.1/1), matching the qualitative pattern reported by Elhage et al. (2022). Increasing the total number of experts removes these sharp boundaries: MoE panels, especially with  $E = 2$ , sometimes resemble their dense counterparts (Figure 4.X.2/2), but the transitions become visibly more continuous. Across settings, at least one expert shifts toward a more monosemantic representation of the relatively important feature. Experts that depart most strongly from the dense phase diagrams also have substantially lower superposition scores, visible as bluer regions. This is consistent with Figure 3: relative to dense baselines, MoEs favor lower-interference representations as network sparsity increases.

For the  $n = 2, m = 1$  condition (Figure 4.A), the dense model ignores the last feature at low feature sparsity, whereas the comparable MoE preserves that feature more often because routing provides an additional allocation mechanism. With three input dimensions and one hidden dimension (Figure 4.B), the margin changes: experts already use superposition for

the other two features, leaving insufficient local capacity for the third feature. Even in this capacity-limited setting, the MoE does not recover the sharp dense phase boundary, except for a partly discrete, mixed region around feature sparsity 0.5–0.7 in 4.B.2/2. For  $m = 2$ , the dense baseline in Figure 4.C.1/1 ignores the last feature in the mid- to low-feature-sparsity region when that feature is relatively less important. As network sparsity increases across the remaining Figure 4.C panels, the last feature is represented with larger L2 magnitude ( $\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$ ). Thus, in the low-relative-importance region, increasing network sparsity raises feature dimensionality, matching the trend in Figure 3.

We also observe an intermediate feature-sparsity band, roughly 0.48 to 0.7, in Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2. In this band, polysemanticity, monosemanticity, and feature ignorance all appear across nearby cells, suggesting that MoEs have a loss-comparable middle regime where several representational strategies are viable. Dense models do not show the same mixed band, which strengthens the empirical conclusion that routed experts learn different allocation strategies rather than reproducing dense phase behavior.

**Conclusion.** All phase-change experiments use top- $k = 1$ , so each input activates only one expert. Even under this restrictive condition, the  $E = 2$  MoEs already differ sharply from dense baselines, including cases where hidden-dimensional capacity with superposition would be sufficient to represent all features. The empirical comparison therefore argues against treating MoEs as simple aggregates of dense models. Routing changes each expert’s observed feature domain, and that change measurably alters both phase behavior and superposition scores.

## 6 EXPERT SPECIALIZATION

Since MoEs exhibit lower superposition in the preceding comparisons, we next ask whether the cleaner feature representations are organized within experts in a way that corresponds to specialization.

Expert specialization in MoEs is often operationalized through load balancing across all inputs (Chaudhari et al., 2025). Load balancing is useful, but it is an aggregate usage metric: an evenly used expert is not necessarily specialized. A more direct empirical criterion is conditional usage—whether an expert is selected more often when the features it represents are active.

We therefore define an expert as specialized when it both *occupies* particular feature directions in input space and represents those same features relatively monosemantically. This definition links routing and representation with measurable quantities. We show that the two conditions are correlated, and that when both hold, expert usage becomes more balanced as a consequence of feature-aligned routing rather than as the definition of specialization itself.

Because we fix  $k = 1$ , the router partitions feature space into convex cone regions (Appendix A.3), with each region assigned to one expert. By definition,  $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$ , where  $C$  is a cone region and  $s$  is any positive scalar. Thus, if a feature vector  $\vec{x}$  is routed to an expert, every scaled version  $s\vec{x}$  is routed to that same expert. We say that feature  $x$  is contained within expert  $e$ , or equivalently that expert  $e$  *occupies*  $x$ .

In small models, distributing the input space across more experts is associated with lower loss (Appendix A.4). This empirical pattern motivates a representation-level question: when an expert receives a particular region of input space, does it also represent the corresponding features more cleanly? Our specialization definition predicts a positive association between occupation and monosemanticity, and we test that association in larger toy models such as  $m = 10$ .

For models with  $n > 2$ , we evaluate the occupation–representation relationship in both directions. First, we test whether initializing experts to occupy specific input features makes those experts more monosemantic with respect to those features. Second, for features that an expert ultimately represents monosemantically, we measure whether that expert also occupies the corresponding input directions.

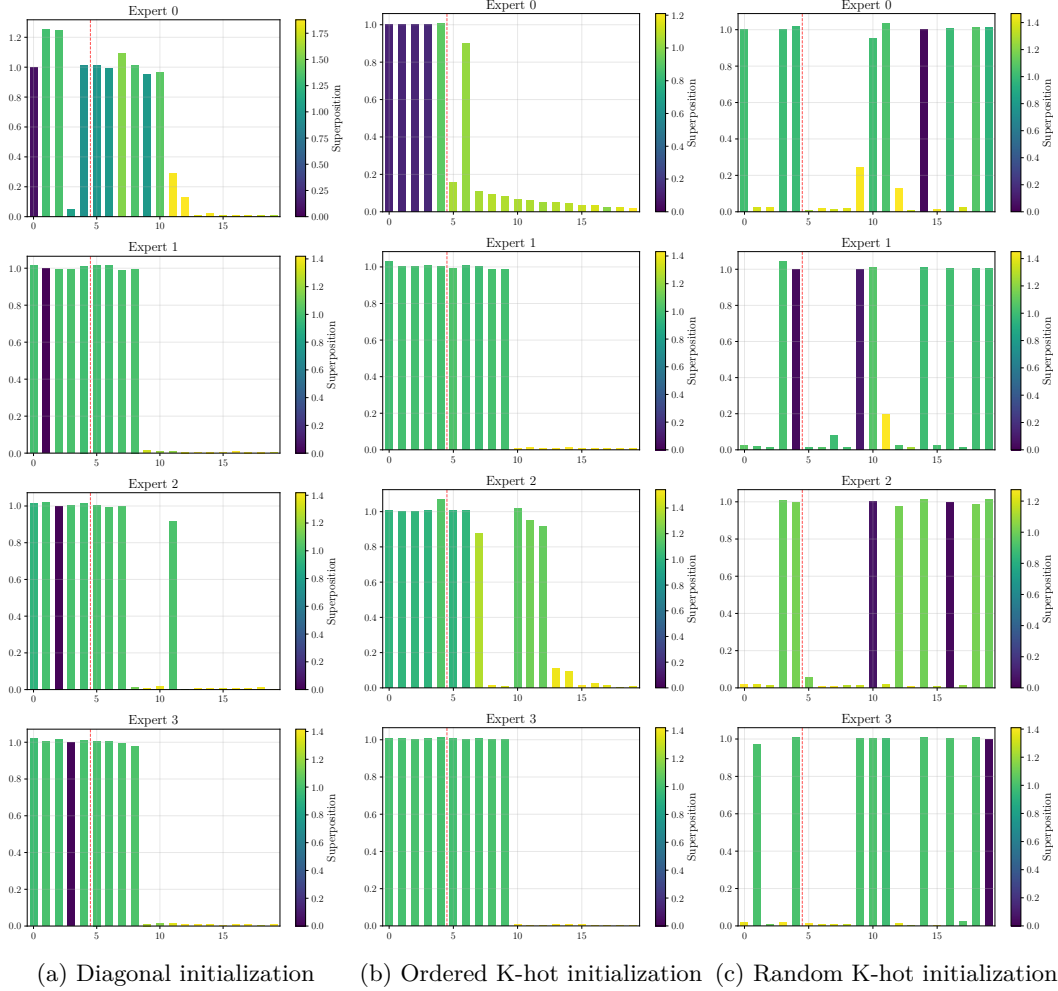


Figure 5: Expert feature norms  $\|W_i^{(e)}\|$  and superposition scores (color) under three router-initialization comparisons, with  $n = 20, m = 5, E = 4, S = 0.1$ . In (a), the gate matrix is initialized along the main diagonal ( $W_i^T = \hat{e}_i$ , the basis vector for that dimension), and relative feature importance decreases exponentially from feature one to feature 20. In (b), the gate matrix uses “ordered k-hot” initialization: the first expert aligns with the first five features, and each subsequent expert aligns with the next five features, with the same feature-importance ordering as (a). In (c), the gate matrix uses “random k-hot” initialization, assigning five random non-overlapping features to each expert so that all 20 features are covered collectively; feature importance still decreases exponentially but is randomly distributed across features.

With diagonal gate initialization, each expert monosemantically represents exactly the single feature it initially occupies, as shown in Figure 5a. With ordered k-hot initialization, the first expert represents four of its five initially occupied features monosemantically (Figure 5b); the other experts, which are initialized on less important feature groups, do not acquire monosemantic representations of those lower-importance features or of the five highest-importance features they were not initialized to occupy. When feature-importance order is broken through random k-hot initialization, each expert monosemantically represents only the most important feature or features within its initially occupied set (Figure 5c).

These comparisons show a strong correspondence between initial routing occupation and eventual monosemantic representation. The margin is also importance-sensitive: experts monosemantically represent important features, whether the initialization assigns one im-

portant feature explicitly or assigns a set of candidate features from which the expert learns to represent the most important member monosemantically.

When feature importance is uniform, experts place all features in superposition and interference levels are higher overall. Even in this less favorable condition, initially occupied features remain *relatively* cleaner: their superposition scores are, on average, 1.03 standard deviations below the expert-level mean.

We next measure the reverse direction: when an expert represents certain features monosemantically, does it also occupy those features in input space? For each expert, we compute usage when its monosemantic feature or features are *one of many* active features, and again when they are the *only* active features. The second condition directly estimates occupation probability. Table 1 shows that the correlation holds under both xavier and k-hot initialization. Because  $E = 10$ , mean usage near  $\sim 10\%$  corresponds to even load balancing; conditional on the expert’s monosemantic feature(s) being active, usage rises well above this baseline in every nonempty group, and when only those feature(s) are active, the expert typically dominates selection. Under k-hot initialization, the alignment is exact by this metric: *100%* of all features monosemantically represented by an expert are also occupied by that same expert.

Table 1: Conditional expert-usage evidence for specialization with  $n = 100, m = 10, E = 10$ . For each initialization scheme (xavier and k-hot), we train 100 models, yielding 1000 experts. Experts are grouped by the number of features they represent monosemantically, and usage is reported under three conditions: average usage across inputs, usage when the expert’s monosemantic feature(s) are active among other features, and usage when those feature(s) are the *only* active features. The comparison shows how conditional usage rises relative to the  $\sim 10\%$  load-balancing baseline when represented features are present.

| Xavier Initialization                      |                                 |                       |                                          |                                               |
|--------------------------------------------|---------------------------------|-----------------------|------------------------------------------|-----------------------------------------------|
| Number of monosemantic features per expert | Number of experts (out of 1000) | Mean expert usage (%) | Mean expert usage; feature(s) active (%) | Mean expert usage; only feature(s) active (%) |
| 0                                          | 461                             | —                     | —                                        | —                                             |
| 1                                          | 387                             | 9.595                 | 17.94                                    | 67.18                                         |
| 2                                          | 138                             | 9.599                 | 30.29                                    | 95.65                                         |
| 3                                          | 13                              | 8.363                 | 40.19                                    | 100.0                                         |
| 4                                          | 1                               | 1.428                 | 14.69                                    | 100.0                                         |
| 5                                          | 0                               | —                     | —                                        | —                                             |
| K-Hot Initialization                       |                                 |                       |                                          |                                               |
| Number of monosemantic features per expert | Number of experts (out of 1000) | Mean expert usage (%) | Mean expert usage; feature(s) active (%) | Mean expert usage; only feature(s) active (%) |
| 0                                          | 335                             | —                     | —                                        | —                                             |
| 1                                          | 382                             | 10.00                 | 23.94                                    | 100.0                                         |
| 2                                          | 227                             | 10.02                 | 46.61                                    | 100.0                                         |
| 3                                          | 47                              | 10.09                 | 62.00                                    | 100.0                                         |
| 4                                          | 8                               | 9.95                  | 70.30                                    | 100.0                                         |
| 5                                          | 1                               | 9.62                  | 74.79                                    | 100.0                                         |

Experts that represent more features monosemantically are therefore more specialized under our definition. Their unconditional usage can remain near or below the load-balancing baseline, but their conditional usage increases sharply when their represented features are active. This pattern holds across the table except for the rare xavier-initialized group with four monosemantic features, where utilization drops substantially.

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## 7 CONCLUSION

We compared dense and MoE toy models using explicit superposition, dimensionality, reconstruction-loss, phase-diagram, and conditional-usage metrics. Across these comparisons, MoEs consistently show greater monosemanticity than dense networks: they localize interference within experts, reduce features per dimension as expert count increases, and replace dense-style discontinuous phase changes with smoother expert-specific transitions. The dense baseline retains a reconstruction-loss advantage, with fixed-sparsity gaps of roughly 0.03–0.08 in our equal-total-parameter experiments, but higher-expert MoEs remain comparable while producing cleaner feature representations. We also proposed a feature-based definition of expert specialization and found that experts naturally align with coherent feature combinations when initialization supports that alignment. These findings are based on simple autoencoder toy models with synthetic data, so generalization to large-scale transformers with unknown feature distributions remains open (Appendix A.6). Even with that caveat, the quantitative pattern is favorable: in these controlled settings, MoEs trade a modest loss margin for substantially more interpretable representational structure. Future work should identify which training and architectural conditions most reliably increase MoE monosemanticity, how MoE training dynamics differ from dense dynamics, and when feature-aligned specialization emerges at scale.

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## A APPENDIX

### A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

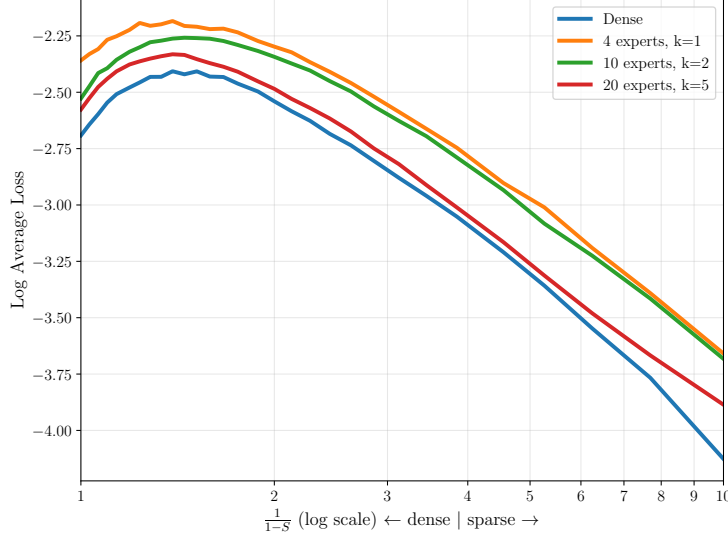


Figure 6: Log average loss versus feature density ( $\frac{1}{1-s}$ ) for the **dense** ( $m = 20$ ) baseline and three equal-total-parameter MoE comparisons: **MoE** (4 experts,  $k = 1$ ,  $m = 5$ ), **MoE** (10 experts,  $k = 2$ ,  $m = 2$ ), and **MoE** (20 experts,  $k = 5$ ,  $m = 1$ ). All models use uniform feature importance ( $I_i = 1.0$ ) for  $n = 100$  input features, and each point averages five runs per sparsity level. The **dense** model has lower loss at every sparsity level, but the MoE curves move closer to the dense baseline as expert count increases.

### A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition exploits the mismatch between sparse feature activations and dense computation by compressing rarely active features into shared dimensions. MoEs address the same mismatch structurally through conditional computation. By aligning activation sparsity with feature sparsity, MoEs reduce the computational penalty of dedicating capacity to rarely active features. As noted by Elhage et al. (2022), when computation is spent only on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes preferable; in our framing, MoEs trade the *compression* of superposition for the *selection* of routing.

Geometrically, this structural change appears as reduced *interference*. Even when the global ratio of represented features to parameters is held fixed, the router partitions feature space so that a feature routed to expert  $e$  competes only with features assigned to the same expert. Interference is therefore governed by the local matrix  $(W^e)^\top W^e$  rather than by the dense model’s global  $W^\top W$ . This partitioning reduces the number of possible interfering feature pairs for any given feature vector, lowering the optimization pressure to pack features in superposition and making monosemantic expert representations more attainable.

### A.3 ROUTER SUBSPACES ARE CONVEX CONES

For  $k = 1$ , the router  $g(x) = \text{softmax}(W^r x)$  selects the same expert as  $\text{argmax}(W^r x)$ . The region routed to expert  $i$  is therefore described by  $\forall j \neq i, (w_i - w_j)^\top x > 0$ , where  $w_i$  and  $w_j$  are row vectors of  $W^r$ . Because this is a homogeneous linear inequality, the routed regions are convex cones. If  $x$  satisfies the inequality, then any positive scaling  $sx$  also satisfies it; if  $x_1$  and  $x_2$  satisfy it, then any convex combination  $x = x_1\lambda + (1 - \lambda)x_2$  for  $\lambda \in [0, 1]$  also satisfies it.

For  $k > 1$ , the set of inputs routed to a particular expert  $e$  becomes a union of convex cones rather than a single cone. Because a union of convex cones is not generally convex, the clean occupation interpretation used in our  $k = 1$  specialization analysis may not transfer directly beyond this routing regime.

#### A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

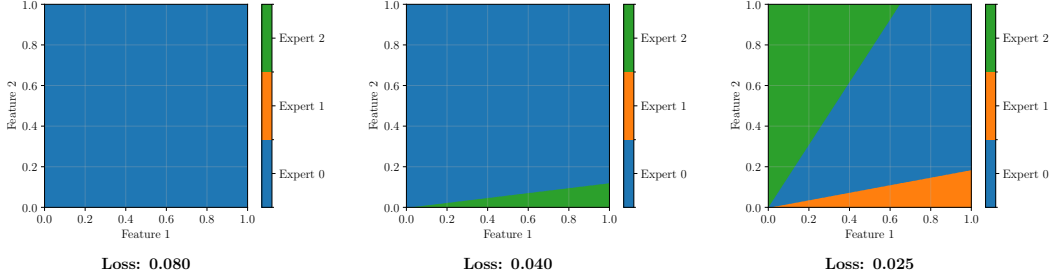


Figure 7: Expert routing for three identical architectures with different initialization outcomes, using  $n = 2$ ,  $E = 3$ , and  $m = 1$ . The left model has the highest loss (0.08) and routes all inputs to one expert. The middle model improves loss to 0.04 by routing a small input region—cases where feature 1 is active—to a second expert. The right model has the lowest loss (0.025) and partitions the input space across all experts: one expert handles feature 1 alone, one handles feature 2 alone, and one handles both active features.

In small models with  $n = 2$ ,  $m = 1$ ,  $E > 1$ , tested for  $E \in [2, 7]$ , lower loss is associated with routing that distributes the input space across more experts. Holding  $n = 2$  makes this comparison visually interpretable: Figure 7 shows which input regions are assigned to which experts and how broader expert coverage coincides with lower reconstruction loss.

#### A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setting of single-layer, single-nonlinearity, top- $k = 1$  MoEs, a theoretical map exists between any dense model and a monosemantic MoE with an equivalent active-feature budget under a sparsity constraint. This construction explains why, when the number of active features is bounded, routing can preserve dense-model functionality while reducing superposition.

Assume that every input has at most  $a$  active features, so  $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$ . Also assume that this active-feature bound does not exceed the hidden dimensionality  $m$  of an expert, and that the expert hidden dimension is no larger than the input dimension:  $a \leq m \leq n$ . To construct the monosemantic MoE, create one expert for each subset  $S \subseteq \{1, 2, \dots, n\}$  with  $|S| \leq a$ , where each expert preserves its assigned subset monosemantically. (Equivalently, it suffices to take only subsets with  $|S| = a$ .) The router then selects the expert corresponding to the observed active-feature set, which contains at most  $a$  features by assumption:

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where  $\text{support}(x) = \{i : x_i \neq 0\}$ . Since  $|S| \leq a \leq m$ , each expert has enough hidden capacity to represent its assigned feature subset without superposition. The construction therefore gives every possible active-feature combination a dedicated monosemantic expert. The number of possible experts required is  $\binom{n}{m}$ .

This theoretical MoE achieves zero reconstruction loss only when the sparsity constraint holds (or in the toy-model limit where sparsity goes to one). If more than  $m$  features are active at once ( $a \not\leq m$ ), the input exceeds the monosemantic capacity of a single expert; a dense polysemantic model faces the same capacity pressure unless the feature distribution

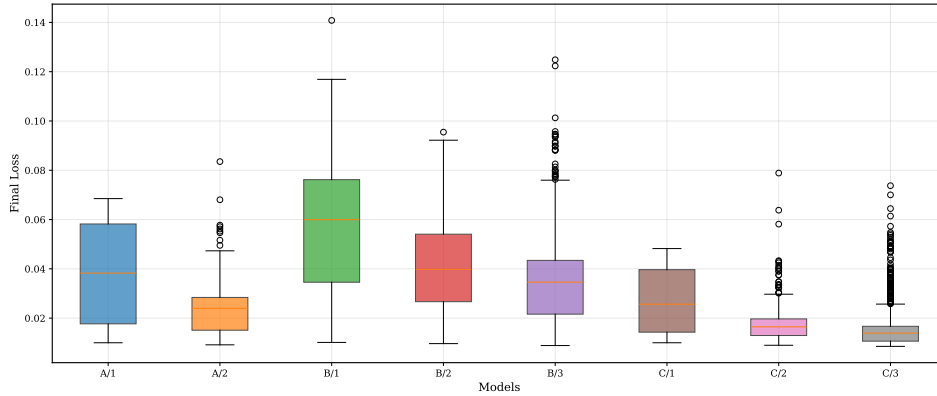


Figure 8: Loss across the same architectures and models used in Figure 4. In label  $X/E$ ,  $X$  denotes the architecture and  $E$  denotes the total number of experts, i.e. network sparsity. Increasing network sparsity lowers mean loss while increasing localized variance, especially when the number of experts reaches the number of input feature dimensions. This variance is consistent with relatively less stable MoE training compared with dense training, even though each cell uses the lowest-loss model among ten runs.

supplies helpful correlations. Thus, even when  $a \not\leq m$  occurs occasionally, the constructed MoE can remain competitive under looser sparsity constraints.

Thus, for any dense model  $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$  under the sparsity constraint  $|\text{support}(x)| \leq a$ , there exists an MoE model  $f_{\text{MoE}}(x)$  such that  $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$  for all valid inputs. In the toy settings studied here, the constraint becomes increasingly accurate as sparsity goes to one. In practice, the number of semantically meaningful active features may be bounded by the information content of the input and by the effective size of its embedding. Under such a bound, a MoE with enough experts and only tractable local superposition may be sufficient to encode the relevant features while remaining more interpretable than a dense polysemantic representation.

## A.6 LIMITATIONS

Our results should be interpreted with several limitations in mind. First, all evidence comes from controlled toy-model settings derived from sparse-autoencoder superposition studies (Elhage et al., 2022). This design is useful because it makes interference, dimensionality, reconstruction loss, and conditional expert usage directly measurable, but it abstracts away many properties of large-scale transformers, including multi-layer computation, attention, heterogeneous feature distributions, and realistic routing dynamics. The transfer of these quantitative patterns to real-world MoE architectures therefore remains uncertain.

Second, our architectural and routing choices are intentionally restricted: we use simple experts, fixed top- $k$  routing (often  $k = 1$ ), controlled active- or total-parameter comparisons between dense and MoE models, and auxiliary load balancing only where stated. Practical MoEs often use richer routing mechanisms, variable expert capacities, and objectives beyond reconstruction, such as next-token prediction (Lepikhin et al., 2021; Fedus et al., 2022). These differences may change both the magnitude and consistency of the monosemanticity margins observed here.

Third, the feature distributions in our synthetic tasks are deliberately simple. Their sparsity patterns, feature-importance schedules, and independence assumptions make comparisons clean, but natural data may introduce correlations and heterogeneity that change when superposition is useful.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under fixed experimental conditions, we do not evaluate downstream task performance, large-scale training stability, or specialization dynamics at scale. Prior work suggests that

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these quantities can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, the study provides favorable mechanistic evidence for MoE monosemanticity under clean experimental conditions, but the reported margins and patterns require validation in larger, realistic MoE systems. Our current measurement methods do not scale naturally to such systems, making scalable superposition and specialization metrics a central direction for future work.